# Project Executable Files

**SMARTBRIDGE - Project Executable Files**

Date: 21/09/2026

Team ID: SWTID-2026-4223

Team Size: 4

Team Leader: Sweatha P

Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K

Project Name: AI FitTrack

Maximum Marks: 3 Marks

## Step 1: Submission Checklist

| S.No | Item to Submit | Submitted (Yes / No / NA) | Remarks / Details |
|---|---|---|---|
| 1 | Complete source code (all files and folders) | Yes | Full Node.js/Express.js backend source code included. |
| 2 | README / Setup Guide (instructions to run) | Yes | Complete README.md file located in the root directory. |
| 3 | requirements.txt / package.json / dependency file | Yes | package.json & package-lock.json included. |
| 4 | Database schema / seed files (if applicable) | Yes | Mongoose Schema files in src/models/User.js & src/models/Workout.js. |
| 5 | Environment configuration file (.env.example) | Yes | .env.example and configured .env included. |
| 6 | Deployed application URL / Executable binary | Yes | Local REST API server at http://localhost:5000. |
| 7 | Dockerfile / Containerization config (if applicable) | NA | Standard Node.js runtime environment used. |
| 8 | APK (if applicable) | NA | Backend RESTful API service. |
| 9 | Test files and test results | Yes | Postman Collection (FitTrack.postman_collection.json) & test logs. |
| 10 | Demo video or walkthrough | Yes | Google Drive walkthrough & API demo video links provided. |

## Step 2: File / Folder Structure

Code Files /

├── node_modules/

```
├── src/
│   ├── config/
│   │   └── db.js # MongoDB connection configuration using Mongoose ODM
│   ├── controllers/
│   │   ├── aiController.js # Handles Gemini AI recommendations and insights logic
│   │   ├── authController.js # User registration, JWT login, and profile controllers
│   │   └── workoutController.js # CRUD operations for workout management
│   ├── middleware/
│   │   ├── auth.js # Express JWT authentication & token verification
│   │   └── errorHandler.js # Centralized error response handling middleware
│   ├── models/
│   │   ├── User.js # Mongoose schema for User collection (with bcrypt)
│   │   └── Workout.js # Mongoose schema for Workout collection
│   ├── routes/
│   │   ├── aiRoutes.js # Express router endpoints for AI services
│   │   ├── authRoutes.js # Express router endpoints for authentication
│   │   ├── index.js # Central route registry
│   │   └── workoutRoutes.js # Express router endpoints for workout CRUD
│   ├── services/
│   │   └── geminiService.js # Google Gemini API (@google/genai) service handler
│   ├── app.js # Express application setup, CORS, and routing middleware
│   └── server.js # Server startup entry point and DB initialization
├── .env # Local environment variables (PORT, MONGO_URI, GEMINI_API_KEY)
├── .env.example # Environment variable blueprint
├── .gitignore # Git ignore file for node_modules and .env
├── FitTrack.postman_collection.json # API collection for Thunder Client / Postman testing
├── package-lock.json # Locked dependency versions
├── package.json # Project dependencies and npm startup scripts
└── README.md # Setup instructions, endpoints guide, and run commands
```

## Step 3: Deployment / Access Details

| Field | Details |
|---|---|
| Hosted / Deployed URL | http://localhost:5000/api |
| Login Credentials (Demo) | Email: sathya@gmail.com<br>Password: Sathya@01 |
| Platform / Hosting Provider | Node.js Express Runtime & MongoDB Atlas Cloud Database |
| Repository Link | https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y |

## Demo Video Links

Demo Video:
https://drive.google.com/file/d/1KswrctSaPI0eOYyJ3GkbbVHvHsU3EfZJ/view?usp=drive_link

API Testing Video:
https://drive.google.com/file/d/1TCXG48a-78ryxF3gipdRMwRBwuhAgPaa/view?usp=drive_link

## Step 4: Run Instructions

Pre-requisites: Node.js v16.0.0 or higher; npm v8.0.0 or higher; Active MongoDB Atlas Cloud Cluster or Local MongoDB; Google Gemini API Key (@google/genai).

Step 1: Open VS Code terminal and navigate into project directory: cd "Code Files"

Step 2: Install dependencies: npm install

Step 3: Configure environment variables in .env file: PORT=5000; MONGO_URI=<your MongoDB Atlas URI>; JWT_SECRET=<your JWT secret>; GEMINI_API_KEY=<your Gemini API key>; GEMINI_MODEL=gemini-2.5-flash

Step 4: Start the REST API application server: npm start

Step 5: Confirm running server status in console: Server running in development mode on port 5000; MongoDB Connected.

Step 6: Open Thunder Client or Postman and test endpoints at http://localhost:5000.

## Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation | Workaround / Status |
|---|---|---|
| 1 | ENOTFOUND MongoDB Connection Delay: Occurs if non-SRV connection strings are used or during internet drops. | Resolved: Upgraded to standard mongodb+srv:// URI format with Mongoose connection retry options. |
| 2 | Port Occupied (EADDRINUSE on Port 5000): Port 5000 remains occupied by an orphaned process. | Resolved: Executed npx kill-port 5000 before running npm start. |
| 3 | Gemini AI Deprecated Model Error (NOT_FOUND): Older model strings fail API calls. | Resolved: Updated model string to gemini-2.5-flash in .env and geminiService.js. |